Assignment 01

① Data base :- digital repository for storing, managing and securing organised collection of data.
- Essential for storing large amount of data volume enabling efficient retrieval, analysis and decision making.
- Examples: financial systems and Healthcare systems.

② Data warehouse: data management system that stores current and historical data from multiple sources in a business friendly manner for easier insight and reporting.
- Difference: Differs from Data base as data warehouse is designed to store, integrate and analyze vast amounts of historical data from multiple sources and database is optimised for recording and updating real time operational data.
- Companies use data warehouse to analyze data.

③ Data lake :- a central location that holds a large amount of data in its native, raw format.
- Data lake stores raw structured or unstructured data for advanced analytics and data warehouse stores processed, structured data for business intelligence and reporting.
- Companies would use data lake for machine learning, predictive analysis, data science exploration and for low cost storage.

④ Data Mart : Subset of data warehouse focused on a particular line of business department of subject area.
- It relates in a way that it is found inside/under warehouse.
- Sales team typically uses it.

(5) Structured data: Highly organised in a fixed format, typically residing in SQL databases e.g. Excel files
Semistructured data: lacks rigid schema but uses tags, markers or hierarchies to define data elements e.g. XML files
Unstructured data: lacks predefined structure or model, resides in native formats - e.g. Email.

(6) Data modeling: process of creating a structured representation of data.
- Important as it reduces data errors, inconsistencies and bridges gap between technical teams and business stakeholders through clear documentation
Types: (1) Physical data model: Represents the actual design of the database, specifying table structures, data types and technical constraints for a particular system.
(2) Logical data model: Adds more detail, such as attributes, primary keys and relationships but remains independent of specific database technology

(7) Data mining: Use of machine learning and statistical analysis to uncover patterns and other valuable information from large data sets.
Uses: to extract actionable insights, predict future trends and uncover hidden ~~trends~~ relationships.
Examples: - Fraud detection in finances through analyzing abnormal transaction patterns.
- Improving patient care in healthcare and predicting outcomes and reducing costs.

(8) Database Management system: software that enables users to create, manage, retrieve and update data in a database, acting as an ~~[illegible]~~ interface between the data and applications.

MySQL : is a relational database management system that stores data in structured tables with rows and columns
- used in websites and apps to store and analyze data

PostgreSQL : a powerful, open source, object-relational database management system that uses and extends SQL language to safely store, manage and scale workloads for web, mobile and analytic applications.
- used in web or mobile apps and acts as primary database.

Oracle : US based information technology company that offers ~~wide~~ wide range of business-oriented products and service that include oracle database
- used in data management to handle complex data.

MongoDB : open source, non-relational database management system that uses flexible documents instead of tables and rows to process and store various forms of data.
- Used in mobile apps.

(9) ETL : data intergration process that retrieves raw data from multiple sources, cleans and converts it into a usable format, and moves it into a central data warehouse or destination for analytics and business intelligence

(A) Extract: Raw data is copied or exported from source locations to a staging area. Structured or unstructured data are extracted from different sources.

(B) Transform - Raw data undergoes data processing. Data is transformed and consolidated for its intended analytical use case

(C) Load : Transformed data is moved from a staging area into a target warehouse. It involves an initial loading of all data, followed by periodic loading of incremental data changes and less often, full refreshes to erase and replace data in the warehouse.

(10) (A) CSV [common Separated values] - a plain text file used to store tabular data, like spreadsheets
- Would use it in bookkeeping

(B) JSON [JavaScript Object Notation] - a lightweight, text based-interchange format used to store and transmit structured data, primarily between a server and a web application
- Would use in configuration files

(C) Parquet: open source column-orientated data file format designed for efficient data storage and retrieval in big data system.
- Would use in high compression data

(D) Avro:- Open source project that provides data serialization and data exchange services for Apache Hadoop.
- Would use in big data Ingestion.

(11) Artificial Intelligence: A branch of computer science that develops system capable of simulating human cognitive functions such as learning, reasoning, problem solving, perception and decision making

(A) Narrow AI: System trained to excel at one particular task or a limited range of tasks - eg. Apples Siri

(B) General AI: Theoretical, highly advanced AI that can understand, learn and apply knowledge similarly to human across a broad range of tasks
eg. Robots

(12) Machine learning: a subset of artificial intelligence that enables systems to learn from data, identifying patterns and make decisions or predictions without explicit programming
Traditional programming - explicitly coded by human programmers.

(A) Supervised learning: Algorithms are trained on labeled data to map inputs to known outputs, commonly used for classification and regression tasks

(B) Unsupervised learning: Algorithms analyze unlabeled data to find hidden structures or patterns such as clustering similar customers

(C) Reinforcement learning: Agents learn to make decisions by taking actions in an environment to maximise rewards

(14) Deep learning: a specialized subset of machine learning driven by multilayered artificial neural networks inspired by human brain.

- It powers advanced AI including generative AI, computer vision and autonomous driving by automatically learning complex patterns from massive unstructured datasets.

Neural networks: computational models inspired by human brain, forming backbone of deep learning and modern AI

- They recognise complex patterns, classifying data and predict outcomes by adjusting weights during training.

(15) Natural language processing is a subfield of artificial intelligence and computer science that enables computers to understand, interpret and generate human language both written and spoken.

(A) Virtual assistance eg Siri

(B) Chat bots - eg - Automated WhatsApp chats.

(C) Translation tools eg. language translation service.

(16) Generative AI: a subset of artificial intelligence that creates new, original contents such as texts, images, code, audio and video by learning patterns from massive database

- It is powered by foundational models and neural networks like transformers

(a) Chat GPT - for writing, coding and brainstorming
(b) Gemini - integrated into google workspace for research and writing
(c) Grok - Realtime AI chatbot with casual tone

(15) large language model: Advance AI systems based on deep learning, especially transformer architectures, trained on massive text databases to understand, summarise, generate and predict human like content
Standard machine learning: subset of artificial intelligence focused on building systems that learn patterns from data to make decisions, predictions or classification without being explicitly programmed for specific rules.
(a) Gemini: Deep integration within the google ecosystems and handling very large context windows.
(b) Grok: designed for real time knowledge access

(17) Training dataset: curated collection of data used to teach machine learning models to identify patterns, make predictions or generate content.
- Important as it acts as the foundation of AI where models analyze input data to adjust internal parameters, enabling them to understand data structures or map inputs to desired output
- It generates skewed, inaccurate or unfair outcomes

(18) Computer vision: A field of AI that enables computers to interpret and understand the visual world, replicating human vision by analyzing digital images and videos
- They analyse numerical pixel values, breaking

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images down into simple patterns (edges, shapes) to identify content.

(a) Medical Imaging: Analyzing images to detect diseases
(b) Autonomous vehicles: Utilizing cameras to identify lanes, obstacles and traffic signs.

(19) Application programming interface: set of protocols and rules that enable different software applications to communicate and exchange data, acting as secure intermediary
- Operates based on predefined rules to securely share only necessary data without exposing an entire system.
- Important as they bridge disconnected systems, enabling the modern digital experiences used daily from mobile apps to complex enterprise cloud architecture

(20) API key: Unique, secret alphanumeric ~~key~~ code used to authenticate and identify an application or user when accessing ~~at~~ an API.
- Can be stored in files, cloud, secrets vaults and git secrets.
- Anyone can access the private data if it is published.

(21) Rest API: Representational State Transfer Application Programming Interface:
- an architectural style for designing network applications that primarily use HTTP to exchange data between client and a server.

(A) GET: Retrieve a resource
(B) POST: Creates a new resource
(C) PUT: Update or replace an existing resource
(D) PATCH: Performs a partial update to a resource.
(E) DELETE: Removes a resource.

(22) JSON. a light weight, text based, language independent data interchange format used for storing and exchanging information.
- Commonly used in web applications for server client communication

eg. { "name" : "Dzudie Mathom"
"age" : 26
"isEmployee" true
"skills" : ["Medicine", "Surgery"]. }

(23) Prompt engineering :- the art of designing, refining and optimizing inputs to guide AI models like LLM to produce accurate, relevant and high quality outputs.
— Important as it involves providing specific context, instructions and examples to improve performance, often acting as a programming in natural language to unlock model capabilities and reduce errors.
(A) "Summarize this article" - "Summarize the key findings of this article in 5 bullets points, focusing on economic for a non-technical audience"
(B) "Go to shop and buy groceries" - "Go to walmart in checkers and buy 5 apples, 3 tomatoes and a pack of guavas".

(24) MAP function : a fundamental tool used in programming, data analysis and hardware control to transform data by applying a specific operation to every element in a collection

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Python: s = ['1', '2', '3', '4']
        res = map(int, s)
        print(list(res))
output: [1, 2, 3, 4]
```

(25) Cloud computing : the on-demanding delivery of IT

resources - including servers, storage, databases and software - over the internet with pay as go pricing.

(1) Iaas (infrastructure - as - a - service) - provides virtualized computing resources like virtual machines, storage and network over the internet.

(2) Paas (Platform - as - a - service) offers platform allowing customers to develop, run and manage applications without building the underlying infrastructure

(3) Saas (Software - as - a - service) Delivers software applications over the internet on a subscription basis

Examples: (1) Iaas - licensing - Ibm cloud
(2) Paas - Google App engine
(3) Saas - microsoft 365

(26) Version Control System: software tool used in software development to track, manage and record changes to source code over time.

Git: most popular distributed version control system, known for speed and flexibility
- used for its speed and flexibility

Commit: snapshot of your files at a specific point in the

Branch: A lightweight, movable pointer to a specific commit

(27) Jupyter notebook: a project to develop open source software, open standards and services for interactive computing across multiple programming language
- Python commonly used
- used to explore data, creating plots and viewing results directly below the code

(28) Python: high level, general purpose programming language that emphasizes code readability, simplicity and

ease of writing with the use of significant indentation plain english naming, an extensive standard library and garbage collection

- Popularity: Due to its unique combination of accessibility, professional utility and user ecosystem

Libraries: ① Numpy: fundamental package for scientific computing providing support for large, multi dimensional arrays and matrices

② Pandas: High performance library for data manipulation and analysis, primarily using its signature "Data frame" structure

③ Scipy: Built on Numpy used for advanced mathematics, scientific and engineering computation

④ PyTorch: flexible deep learning framework favoured in research for its dynamic computation graphs

(29) Data privacy: the right of individuals to control how their personal information is collected, stored and used by organisation

Importancy: Focuses on managing data transparency, consent and security to prevent unauthorised access or misuse.

GDPR: comprehensive European Union law, effective May 25, 2018 designed to protect personal data, privacy and fundamental rights of individuals within EU and EEA

Countries: Applies to 27 European Union countries and 3 European Economic Area countries

(30) Data Ethics: the moral obligations and principles guiding the responsible collection, protection, analysis and usage of data

Unethical use of data:

① Insurance rate Manipulation - potentially violates health privacy
② Consent violations: peoples data are used without their permission
③ Commingling - data is used for a different purpose whilst collected for another different purpose.

(31) Data Governance: discipline that manages the availability, usability, integrity and security of an organisation's data through defined policies, standards and roles.

① Core pillars: Data governance focuses on people and technology. It establishes who can take what action upon what data under what circumstances and using what methods.

② Roles and responsibility: Effective programs include data stewards, a steering committee and data owners, which ensure consistency and accountability.

③ Data lifestyle management: It covers data from creation to deletion including collection, storage and usage

(32) Bias: systematic unfair errors in AI system outputs caused by prejudiced training data, flawed algorithm design or human prejudice

Examples and consequences: ① Facial recognition: often struggles with identifying darker skinned individuals

② Hiring: Can favor male applicants over females due to historical data

③ Language models: May associate specific professions with gender stereotypes

(33)

	Data Analyst	Data Scientist	Data Engineer	AI Engineer
Roles	Analyze existing data to generates insight and support data driven decision making	Develop and implement statistical models and machine learning algorithms to derive insights and make data driven products	Design, develop and deploy machine learning systems to ensure scalability, performance and reliability in production	Develop and deploy generative AI models and applications for content generation, and personalized experiences.
Skills	SQL, Excel Statistics, Data Visualization Python	Python/R machine learning Statistics and probability Data Wrangling Data visualization	Machine learning Modelling Communication & Requirements Gathering, data Analysis, Problem solving Basic SQL & Excel	Python (Transformers, Pytorch, Tensorflow) Hugging Face Lang chain
Tools	SQL, Excel Tableau, Java	Excels Tablea	Spark Hadoop	Lang Sapy LLAMS

(34) Business Intelligence: the procedural and technical infrastructure that collects, stores and analyses data produced by a companies activities

Power BI: a unified, scalable platform for business intelligence data visualization and analytics that connects to vast data sources to create interactive reports and dashboards

Tableau: a leading AI-driven visual analytics platform used to create interactive, drag and drop dashboards and charts from diverse data sources.

(35) JSON and Python: Seeing how people produce

outcomes by coding really is a good thing and motivates me to do better.

36 Cloud computing
- Microsoft Azure

37 I see ~~myself~~ myself as a data Engineer after completing Data Analysis and Data Science and improving my vocabulary and concepts
- Eg:- Coding my own Apps and dashboards
- Making good salary

References

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